

# Intergenerational Housing Misallocation and Fertility

## Compact Theory and Quantitative OLG Blueprint, without Space

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### Abstract

This note develops a non-spatial theory of intergenerational housing mismatch and fertility. Part I gives a compact analytical model with income and liquid-wealth heterogeneity, young fertility choices, old housing retention, financial constraints, a property-tax capitalization channel, a bequest-tax channel, and increasing housing supply. Part II gives a quantitative OLG blueprint with many ages, stochastic income, renters and owners choosing housing size, fixed-rate mortgages, present-value lock-in, estate taxation, bequest transmission, sequential fertility, and construction of family-capable housing. The quantitative block is a model specification and calibration scaffold, not a results section: it reports no calibrated parameters, model fit, transition paths, or counterfactual numbers.

**Executive summary.** Old ownership of large houses is not inefficient by itself. Inefficiency arises when young potential parents are financially constrained and old owners face private retention wedges from mortgage lock-in, tax-basis rules, or policy-created bequest incentives. With increasing housing supply, the planner reallocates existing housing and builds until the shadow value of housing equals marginal construction cost. Bequest-tax repeal is sign-ambiguous: it relaxes heirs' down-payment constraints, but it can also reduce estate-tax rebates, increase old retention, and raise equilibrium housing costs.

**Remark 1** (Status). *The analytical core is closed-form where possible. The quantitative section is intentionally a blueprint. To become a quantitative paper section it needs a parameter vector, data moments, model fit, transition paths, and counterfactual fertility and welfare numbers.*

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## Part I

# Compact Analytical Model

## 1 Environment

Time is discrete. In the compact block there are two adult generations alive in a period: young households  $i$  and old households  $j$ . Young households are fertile and demand family housing; old households are empty nesters and hold part of the existing durable stock. There is no space: all households trade in a single national housing market.

Housing services are produced at convex resource cost

$$C(H; Z), \quad C_H(H; Z) > 0, \quad C_{HH}(H; Z) > 0, \quad C_Z(H; Z) < 0, \quad (1)$$

where  $H$  is aggregate housing services and  $Z$  is construction capacity. Competitive developers satisfy

$$q = C_H(H^S(q; Z); Z), \quad H_q^S(q; Z) > 0, \quad (2)$$

where  $q$  is the flow user cost of housing. The fixed-stock model is the limiting case  $H_q^S = 0$ . This compact block treats  $C$  as a flow-services technology. The durable-investment version used in the quantitative block prices assets instead, with  $P = C'$  and rents pinned down by user-cost arbitrage.

The asset price per unit of housing services is

$$P = \frac{q}{\rho_p}, \quad \rho_p = r + \delta + \tau^p, \quad (3)$$

where  $r$  is the interest rate,  $\delta$  is depreciation, and  $\tau^p$  is a recurring property tax. Equation (3) is the property-tax capitalization channel: for a given  $q$ , higher recurring taxation lowers the purchase price and the required down payment.

Young type  $i$  is

$$i = (y_i, a_i, B_i), \quad (4)$$

where  $y_i > 0$  is labor income,  $a_i \geq 0$  is liquid wealth, and  $B_i \geq 0$  is gross inheritance exposure. Let

$$x = 1 - \tau^b \quad (5)$$

be the after-tax bequest share. Estate-tax revenue is rebated through type-specific transfers  $T_i^b(x)$  satisfying

$$\int T_i^b(x) dF_Y(i) = (1 - x) \int E_j(q, x) dD_O(j), \quad (6)$$

where  $E_j$  is the taxable estate and  $D_O$  is the decedent distribution. Effective liquid wealth is

$$A_i(x) = a_i + xB_i + T_i^b(x). \quad (7)$$

The assignment of  $B_i$  can be family-linked by a kernel  $\Gamma(i | j)$  or anonymous through the cross-sectional inheritance distribution. In either case, gross bequest principal adds up:

$$\int B_i dF_Y(i) = \int E_j(q, x) dD_O(j), \quad (8)$$

with  $B_i$  evaluated at the baseline allocation in the local comparative statics below unless the full inheritance assignment is recomputed. The rebate  $T_i^b$  is the estate-revenue-financed component of the total transfer schedule  $T(s)$  used in the fiscal budget.

Old type  $j$  is

$$j = (m_j, H_j^0, \ell_j, \omega_j^p), \quad (9)$$

where  $m_j > 0$  is old-age income,  $H_j^0$  is initial housing,  $\ell_j \geq 0$  is a mortgage or tax-basis lock-in wedge, and  $\omega_j^p \geq 0$  is a policy-created retention wedge tied to after-tax housing bequests. Define

$$L_j^p(q, x) = \ell_j + \omega_j^p x q, \quad \tilde{q}_j(q, x) = q - L_j^p(q, x), \quad (10)$$

and assume  $\tilde{q}_j(q, x) > 0$ .

**Remark 2** (True bequest preferences versus retention wedges). *The term  $L_j^p$  is a private wedge, not a true bequest preference. It represents a retention subsidy, a tax-basis advantage, or a forgone mortgage benefit that lowers the private cost of keeping housing. If old households instead value bequests directly through warm-glow utility, the planner must count that utility.*

Table 1: Compact-model notation

| Symbol                   | Meaning                                                                   |
|--------------------------|---------------------------------------------------------------------------|
| $q, P$                   | Flow user cost and asset price of housing services                        |
| $H, Z, C(H; Z)$          | Aggregate housing services, construction shifter, construction cost       |
| $y_i, a_i, B_i$          | Young income, liquid wealth, gross inheritance exposure                   |
| $A_i(x)$                 | Liquid wealth after bequest taxation and rebates, $a_i + xB_i + T_i^b(x)$ |
| $h_i, c_i, n_i$          | Young housing, consumption, fertility                                     |
| $s_i = h_i - \kappa n_i$ | Housing slack after children                                              |
| $\chi_i$                 | Non-housing child cost                                                    |
| $\zeta_i$                | Young housing-finance wedge                                               |
| $m_j, H_j^0$             | Old income and initial housing                                            |
| $L_j^p(q, x)$            | Policy-created retention wedge                                            |
| $Q$                      | Planner's shadow value of housing services                                |

## 2 Young households

Young household  $i$  chooses consumption  $c_i$ , housing  $h_i$ , and continuous fertility  $n_i > 0$ . Children require housing. Let

$$s_i = h_i - \kappa n_i > 0, \quad (11)$$

where  $\kappa > 0$  is housing required per child. Preferences are

$$u_i^Y(c_i, h_i, n_i) = \log c_i + \alpha \log(h_i - \kappa n_i) + \beta \log n_i, \quad \alpha, \beta > 0. \quad (12)$$

The non-housing child cost is

$$\chi_i = \chi_0 + \xi y_i, \quad \chi_0 > 0, \quad \xi \geq 0. \quad (13)$$

The young household solves

$$\max_{c_i, h_i, n_i} \log c_i + \alpha \log(h_i - \kappa n_i) + \beta \log n_i \quad (14)$$

subject to

$$c_i + qh_i + \chi_i n_i = y_i, \quad (15)$$

$$(1 - \phi)Ph_i \leq A_i(x), \quad \phi \in (0, 1), \quad (16)$$

$$qh_i \leq \psi y_i, \quad \psi \in (0, 1). \quad (17)$$

This is a collateral-constrained user-cost representation. The flow budget prices housing services at  $q$ , while the down-payment constraint uses the capitalized asset price  $P$  to measure the liquid collateral capacity required to occupy owner housing. Liquid wealth  $A_i$  is not a goods expenditure in this static block; the quantitative model below tracks equity and debt explicitly. Accessible housing is

$$h_i \leq \bar{h}_i(q, \tau^P, x) = \min \left\{ \frac{A_i(x)\rho_P}{(1 - \phi)q}, \frac{\psi y_i}{q} \right\}. \quad (18)$$

Let  $\lambda_i$ ,  $\mu_i$ , and  $\eta_i$  be multipliers on (15), (16), and (17). The FOCs are

$$\frac{1}{c_i} = \lambda_i, \quad (19)$$

$$\frac{\alpha}{h_i - \kappa n_i} = \lambda_i q + \mu_i(1 - \phi)P + \eta_i q, \quad (20)$$

$$\frac{\beta}{n_i} = \lambda_i \chi_i + \frac{\alpha \kappa}{h_i - \kappa n_i}. \quad (21)$$

Define the goods-equivalent housing-finance wedge

$$\zeta_i = \frac{\mu_i}{\lambda_i}(1 - \phi)P + \frac{\eta_i}{\lambda_i}q \geq 0. \quad (22)$$

Then

$$\frac{\alpha c_i}{h_i - \kappa n_i} = q + \zeta_i. \quad (23)$$

Combining (21) and (23), fertility satisfies

$$\boxed{\frac{\beta c_i}{n_i} = \chi_i + \kappa(q + \zeta_i)}. \quad (24)$$

Thus a binding financial constraint acts like an implicit child tax  $\kappa \zeta_i$ . Because  $\log n_i$  imposes  $n_i > 0$ , the compact model studies the intensive fertility margin. The discrete sequential block below restores childlessness and parity transitions.

If  $\zeta_i = 0$ , define  $\mathcal{A} = 1 + \alpha + \beta$ . The closed-form demands are

$$c_i^u = \frac{y_i}{\mathcal{A}}, \quad s_i^u = \frac{\alpha y_i}{\mathcal{A}q}, \quad (25)$$

$$n_i^u = \frac{\beta y_i}{\mathcal{A}(\chi_i + \kappa q)}, \quad h_i^u = s_i^u + \kappa n_i^u. \quad (26)$$

If the housing cap binds at  $h_i = \bar{h}_i$ , define  $Y_i^c = y_i - q\bar{h}_i$ . Fertility solves

$$\max_{n_i} \log(Y_i^c - \chi_i n_i) + \alpha \log(\bar{h}_i - \kappa n_i) + \beta \log n_i. \quad (27)$$

Let

$$M_i = \kappa Y_i^c(\alpha + \beta) + \chi_i \bar{h}_i(1 + \beta). \quad (28)$$

The interior constrained fertility choice is

$$n_i^c = \frac{M_i - [M_i^2 - 4\kappa\chi_i(1 + \alpha + \beta)\beta Y_i^c \bar{h}_i]^{1/2}}{2\kappa\chi_i(1 + \alpha + \beta)}. \quad (29)$$

In the housing-bottleneck region,

$$\frac{\partial n_i^c}{\partial \bar{h}_i} > 0 \quad \text{if} \quad \frac{\alpha\kappa}{(\bar{h}_i - \kappa n_i)^2} > \frac{\chi_i q}{(Y_i^c - \chi_i n_i)^2}. \quad (30)$$

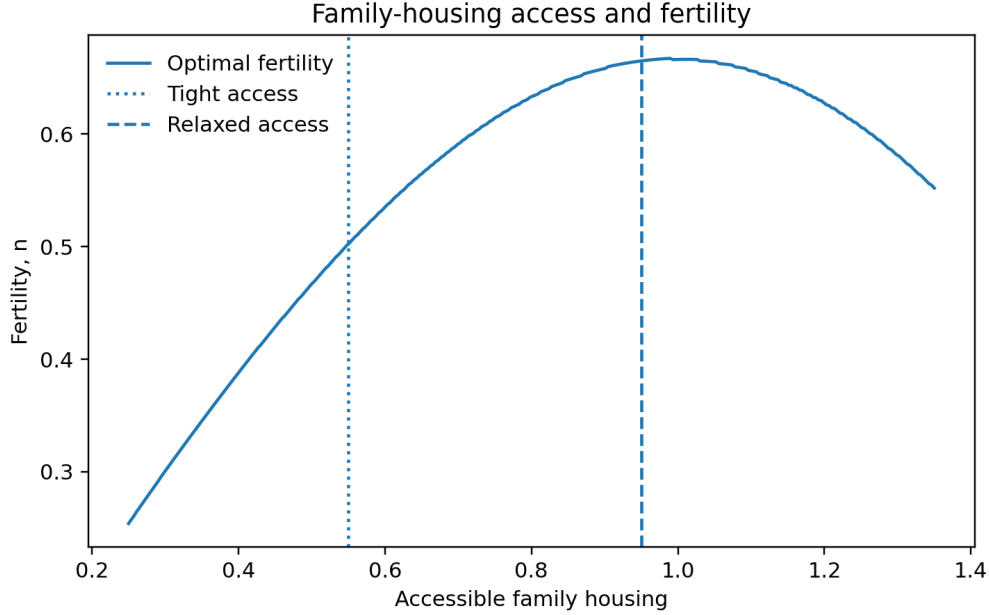


Figure 1: Schematic: family-housing access and fertility. In the bottleneck region, accessible housing raises fertility by reducing crowding per child.

### 3 Old households and fiscal counterpart

Old household  $j$  chooses consumption  $c_j^O$  and retained housing  $h_j^O$ . The compact problem is

$$\max_{c_j^O, h_j^O} \log c_j^O + \gamma \log h_j^O \quad (31)$$

subject to

$$c_j^O + \tilde{q}_j(q, x)h_j^O = m_j + qH_j^0, \quad 0 \leq h_j^O \leq H_j^0. \quad (32)$$

The right-hand side values the initial endowment at social opportunity cost  $q$ . The left-hand side charges the private retention cost  $q - L_j^p$ . Equivalently, the old household sells the endowment at  $q$ , buys back retained housing at  $q$ , and receives private benefit  $L_j^p$  per retained unit. This is reduced-form accounting and requires a fiscal, lender, or tax-expenditure counterpart.

For an interior solution,

$$\frac{\gamma c_j^O}{h_j^O} = q - L_j^p(q, x), \quad (33)$$

so

$$h_j^O(q, x) = \frac{\gamma}{1 + \gamma} \frac{m_j + qH_j^0}{q - L_j^p(q, x)} \quad (34)$$

subject to  $h_j^O \leq H_j^0$ .

The aggregate cost of policy-created retention benefits is

$$\mathcal{S}^L(q, x) = \int L_j^p(q, x) h_j^O(q, x) dF_O(j). \quad (35)$$

If the wedge is fiscal, the compact government budget is

$$\tau^p PH + (1 - x) \int E_j dD_O(j) = \int T(s) dF(s) + G + \mathcal{S}^L(q, x). \quad (36)$$

Here  $T(s)$  is total household transfers and includes  $T^b(s)$  as its estate-revenue-financed component; (6) is therefore an incidence restriction, not an additional use of revenue. If the wedge is a low-coupon mortgage benefit, (36) is replaced by a lender-borrower transfer account. Transfers affect incidence and welfare weights, but they do not relax the aggregate resource constraint unless they involve deadweight costs.

## 4 Equilibrium and characterization

Let  $F_Y$  and  $F_O$  be the distributions of young and old types. Aggregate housing demand is

$$H^D(q; \tau^p, x) = \int h_i^Y(q, \tau^p, x) dF_Y(i) + \int h_j^O(q, x) dF_O(j). \quad (37)$$

**Definition 1** (Competitive equilibrium). *Given policies  $(\tau^p, \tau^b, Z)$ , a competitive equilibrium is a user cost  $q$ , asset price  $P$ , young allocations  $\{c_i, h_i, n_i\}$ , old allocations  $\{c_j^O, h_j^O\}$ , and supply  $H$  such that households optimize, developers optimize,  $P = q/(r + \delta + \tau^p)$ , fiscal and lender accounts are closed, and*

$$H^D(q; \tau^p, x) = H^S(q; Z). \quad (38)$$

Aggregate fertility is  $N(q; \tau^p, x) = \int n_i(q, \tau^p, x) dF_Y(i)$ .

**Proposition 1** (Existence and uniqueness). *Suppose  $H^D$  is continuous and strictly decreasing in  $q$ ,  $H^S$  is continuous and weakly increasing in  $q$ , and the curves cross. For old demand, impose  $\tilde{q}_j(q, x) > 0$  and  $1 - \omega_j^p x > 0$ , so (34) is decreasing in  $q$  on interior regions. Then the compact economy has a unique equilibrium user cost  $q^e$ .*

*Proof.* Market clearing is the scalar equation  $H^D(q) = H^S(q)$ . Continuity and crossing imply existence. Since  $H_q^D - H_q^S < 0$ , the crossing is unique.  $\square$

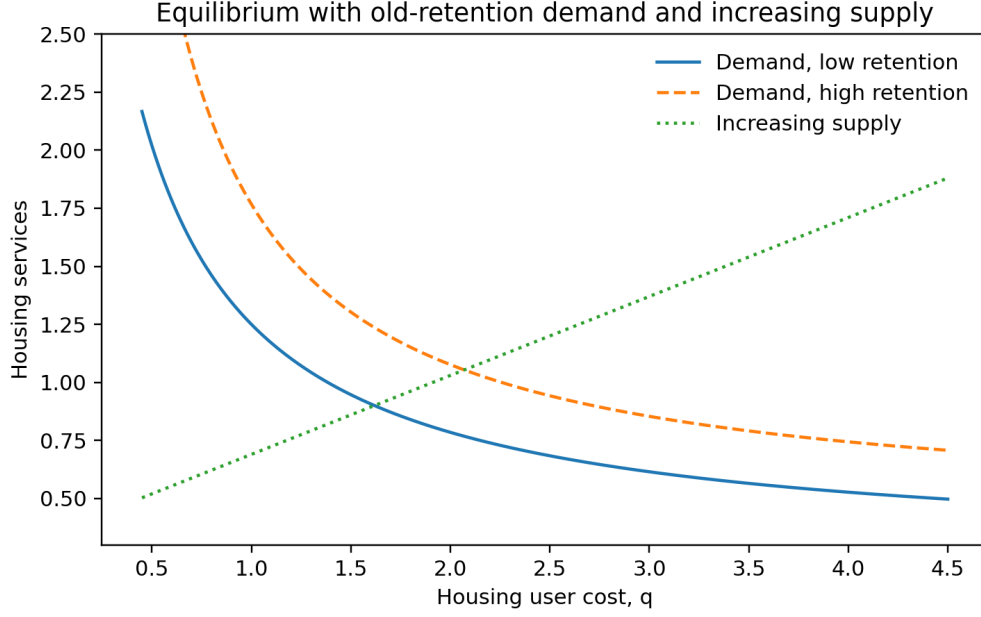


Figure 2: Schematic: equilibrium with increasing housing supply. Old lock-in shifts demand outward; supply elasticity attenuates the price response.

The young marginal value of housing is

$$MV_i^Y = \frac{\alpha c_i}{h_i - \kappa n_i} = q + \zeta_i. \quad (39)$$

The old private marginal value condition is

$$MV_j^O = \frac{\gamma c_j^O}{h_j^O} = q - L_j^p(q, x). \quad (40)$$

Therefore

$$\boxed{MV_i^Y - MV_j^O = \zeta_i + L_j^p(q, x).} \quad (41)$$

Income heterogeneity alone is not an inefficiency. The inefficiency comes from constrained young demand and private old-retention wedges.

## 5 Planner

The constrained planner chooses allocations and aggregate housing subject to

$$\int h_i^Y dF_Y(i) + \int h_j^O dF_O(j) \leq H, \quad (42)$$

$$\int (c_i + \chi_i n_i) dF_Y(i) + \int c_j^O dF_O(j) + C(H; Z) \leq Y, \quad (43)$$

where  $Y = \int y_i dF_Y(i) + \int m_j dF_O(j)$ . Let  $Q$  be the shadow value of housing and  $\nu_i^g \geq 0$  the goods-equivalent social value of an additional birth. Interior FOCs are

$$\frac{\alpha c_i}{h_i - \kappa n_i} = Q, \quad (44)$$

$$\frac{\gamma c_j^O}{h_j^O} = Q, \quad (45)$$

$$Q = C_H(H; Z), \quad (46)$$

$$\frac{\beta c_i}{n_i} + \nu_i^g = \chi_i + \kappa Q. \quad (47)$$

The planner reallocates across generations and builds until the shadow value of housing equals marginal construction cost. The baseline model sets  $\nu_i^g = 0$ . A positive  $\nu_i^g$  is an optional exogenous social valuation of births, for example from a fiscal or demographic motive not micro-founded in this draft.

**Proposition 2** (Marginal misallocation). *Consider transferring one marginal unit of housing from old type  $j$  to young type  $i$ , with compensating goods transfers. Let  $R_j \geq 0$  be any real preference or resource cost of reducing old housing. At the competitive allocation, the goods-equivalent social gain is*

$$\Delta_{ij} = \zeta_i + L_j^p(q, x) + \nu_i^g \frac{\partial n_i}{\partial h_i} - R_j. \quad (48)$$

The allocation is locally constrained inefficient on margin  $(i, j)$  iff  $\Delta_{ij} > 0$ .

*Proof.* The young-old private value gap is  $(q + \zeta_i) - (q - L_j^p) = \zeta_i + L_j^p$ . Add the social fertility value generated by extra young housing and subtract any real old-age cost the planner counts. True old housing utility and true bequest utility enter preferences; they are not wedges.  $\square$

## 6 Policy incidence

**Property tax.** For a wealth-constrained young household,

$$h_i = \frac{A_i(x)(r + \delta + \tau^p)}{(1 - \phi)q}. \quad (49)$$

Hence

$$d \log h_i = \frac{d\tau^p}{r + \delta + \tau^p} - d \log q. \quad (50)$$

A property-tax increase raises accessible housing iff

$$\frac{\partial \log q^e}{\partial \tau^p} < \frac{1}{r + \delta + \tau^p}. \quad (51)$$

For an income-constrained household,  $h_i = \psi y_i / q$ , so the sign is  $-d \log q$ .

**Supply expansion.** A construction reform is  $Z \uparrow$ . Differentiating market clearing gives

$$\frac{dq^e}{dZ} = \frac{H_Z^S}{H_q^D - H_q^S} < 0. \quad (52)$$

Supply expansion lowers user costs and relaxes both collateral and flow-affordability constraints.

**Bequest-tax repeal.** Repeal is  $x \uparrow$ . From (7),

$$\frac{\partial A_i}{\partial x} = B_i + T_{i,x}^b. \quad (53)$$

Differentiating (6) gives the aggregate rebate restriction

$$\int T_{i,x}^b dF_Y(i) = - \int E_j dD_O(j) + (1-x) \frac{\partial}{\partial x} \int E_j(q, x) dD_O(j). \quad (54)$$

For revenue-neutral repeal,  $T_{i,x}^b < 0$  for rebate recipients. For deficit-financed repeal,  $T_{i,x}^b = 0$  in the repeal period. Market clearing implies

$$\frac{dq^e}{dx} = - \frac{H_x^D - H_x^S}{H_q^D - H_q^S}. \quad (55)$$

In the baseline bequest experiment  $H_x^S = 0$  unless estate-tax policy is also allowed to shift construction or developer costs. For a wealth-constrained young type,

$$\boxed{\frac{d \log h_i}{dx} = \frac{B_i + T_{i,x}^b}{A_i(x)} - \frac{d \log q^e}{dx}}. \quad (56)$$

Repeal raises this type's housing iff the direct net-liquidity effect exceeds the equilibrium user-cost effect.

**Proposition 3** (Bequest-tax repeal is ambiguous). *Bequest-tax repeal raises aggregate fertility only if the direct net-liquidity gains to constrained heirs dominate the loss of rebates, old-retention effects, and equilibrium housing-cost effects. The sign is more likely positive when inheritances accrue to high-income, low-liquid-wealth young households near the fertility margin. The sign is more likely negative when repeal mainly raises old retention, removes transfers from constrained households, or raises prices of family-capable housing.*

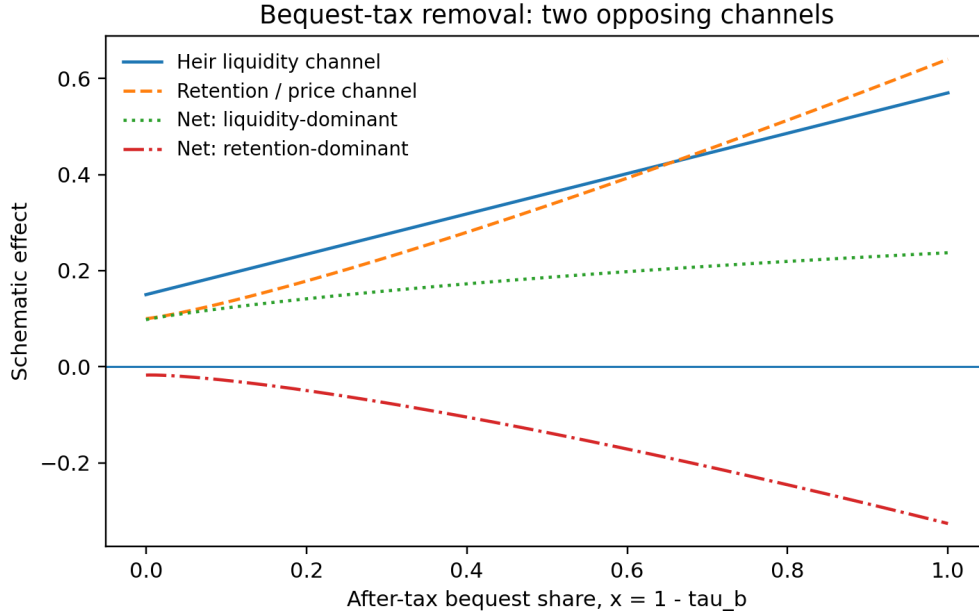


Figure 3: Schematic: bequest-tax repeal has ambiguous fertility effects. Repeal relaxes heirs' down-payment constraints but can also increase old retention and raise prices.

## Part II

# Quantitative OLG Blueprint without Space

## 7 Purpose, timing, and state space

The quantitative block embeds the compact wedge in a finite-life OLG economy with income risk, tenure and house-size choice, mortgages, bequests, sequential fertility, and increasing supply. It remains non-spatial. It is not a results section.

Time is annual. Households live from age  $a = 1$  to  $A$ . Survival is  $\pi_a$ . Fertility is possible for  $a \in \mathcal{F}$ . The household chooses a birth indicator

$$f_t \in \{0, 1\}, \quad n_{t+1} = n_t + f_t \leq \bar{n}. \quad (57)$$

Children are tracked by a child-age vector  $g_t = (g_{0,t}, \dots, g_{S_c,t})$  with law of motion

$$g_{t+1} = (f_t, g_{0,t}, \dots, g_{S_c-1,t}), \quad m(g_t) = \sum_{s=0}^{S_c} g_{s,t}. \quad (58)$$

This fixes fertility timing as sequential parity hazards; there is no one-shot completed-fertility jump.

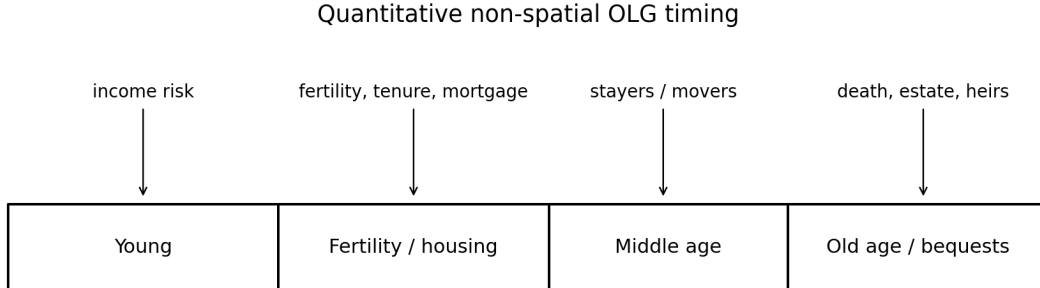


Figure 4: Schematic: quantitative non-spatial OLG timing. Households move through income risk, sequential births, housing adjustment, mortgage choices, aging, death, and bequest transmission.

The individual state is

$$s_t = (a, z, b, o, k, d, m^d, \iota, n, g, B), \quad (59)$$

where  $a$  is age,  $z$  is idiosyncratic productivity,  $b$  is liquid assets,  $o \in \{R, O\}$  is tenure,  $k \in \mathcal{K}$  is occupied housing size,  $d$  is mortgage debt,  $m^d$  is remaining mortgage maturity,  $\iota$  is the incumbent coupon,  $n$  is completed fertility,  $g$  is the child-age vector, and  $B$  is inherited liquid wealth.

The key convention is

$$\mathcal{K} = \{S, M, L\}. \quad (60)$$

The size  $k$  produces housing services for both renters and owners. Tenure  $o$  determines whether the household pays rent or owns. Renters have  $d = 0$ ,  $m^d = 0$ , and no incumbent coupon, but they still choose  $k$  and consume  $h_k$  services.

Income is

$$y_t(a, z) = W_t e_a z, \quad z' \sim \Pi_z(\cdot | z). \quad (61)$$

Table 2: Quantitative state variables and units

| Object                            | Meaning                                                |
|-----------------------------------|--------------------------------------------------------|
| $a, z, b$                         | Age, idiosyncratic productivity, liquid assets         |
| $o \in \{R, O\}$                  | Tenure: rent or own                                    |
| $k \in \mathcal{K} = \{S, M, L\}$ | Occupied housing size for both renters and owners      |
| $h_k$                             | Service units delivered by one size- $k$ dwelling      |
| $d, m^d, \iota$                   | Mortgage balance, remaining maturity, incumbent coupon |
| $n, g$                            | Completed fertility and child-age vector               |
| $B$                               | Inherited liquid wealth                                |
| $P_{k,t}$                         | Asset price per service unit of size- $k$ housing      |
| $R_{k,t}$                         | Rental/user cost per service unit of size- $k$ housing |
| $H_{k,t}, I_{k,t}$                | Stock and construction of service units of size $k$    |

## 8 Preferences and housing services

Housing slack is

$$\sigma(k, g) = h_k - \kappa m(g). \quad (62)$$

Period utility is

$$u(c, k, n, g) = \frac{[c^{1-\alpha_h} \sigma(k, g)^{\alpha_h}]^{1-\sigma_u} - 1}{1 - \sigma_u} + \varphi_n v(n), \quad (63)$$

with the log limit when  $\sigma_u = 1$ . The non-housing child cost is

$$\chi_t(a, z, g) = \chi_0 m(g) + \xi y_t(a, z) m(g). \quad (64)$$

## 9 Tenure, mortgages, and lock-in

A renter of size  $k'$  pays  $R_{k',t} h_{k'}$  and chooses  $d' = 0$ . An owner who buys size  $k'$  pays  $P_{k',t} h_{k'}$  and can borrow  $d'$ . Since  $P_{k,t}$  is a price per service unit,  $P_{k,t} h_k$  is the asset price of a size- $k$  dwelling. New owner mortgages satisfy

$$d' \leq \phi P_{k',t} h_{k'}, \quad (65)$$

$$\mathcal{P}(d', R_t^m, m_0) \leq \psi y_t(a, z), \quad (66)$$

where  $m_0$  is origination maturity. The fixed annual mortgage payment is

$$\mathcal{P}(d, R, m) = d \frac{R}{1 - (1 + R)^{-m}}, \quad (67)$$

with zero-rate limit  $d/m$ .

An incumbent owner keeping the same house keeps the old coupon  $\iota$ . Moving or refinancing replaces it with  $R_t^m$ . The present-value lock-in cost of losing a below-market coupon is

$$\mathcal{L}_t^m(d, \iota, m^d; R_t^m) = \max \left\{ \sum_{s=1}^{m^d} \frac{\mathcal{P}(d, R_t^m, m^d) - \mathcal{P}(d, \iota, m^d)}{(1 + r_t)^s}, 0 \right\}. \quad (68)$$

This is a household cash-flow measure of lock-in, discounted at the risk-free rate. A market-value liability measure would discount the remaining mortgage cash flows at mortgage-market rates; the present object is chosen to isolate the payment wedge relevant for moving decisions. Adjustment cost is

$$\Phi_t(k, d, m^d, \iota) = \lambda_s P_{k,t} h_k + \mathcal{L}_t^m(d, \iota, m^d; R_t^m). \quad (69)$$

The real resource cost is  $\lambda_s P_{k,t} h_k$ . The coupon term is a borrower-lender transfer unless mortgage investors are outside the economy.

The budget is written compactly as

$$c + b' + \mathcal{X}_t(o', k', d', m^d, \iota' \mid o, k, d, m^d, \iota) + \chi_t(a, z, g') = y_t(a, z) + (1 + r_t)b + T_t(s), \quad (70)$$

where  $\mathcal{X}_t$  is evaluated by tenure branch: renters pay rent; owner stayers pay mortgage payments, maintenance, and property taxes; movers sell, repay debt, forfeit the coupon value, pay transaction costs, buy, and choose new debt, maturity, and coupon; sellers liquidate equity and rent.

## 10 Bequests, estate taxation, and transfers

At death, the estate is

$$E_t(s) = \max\{b + \mathbf{1}\{o = O\}(P_{k,t} h_k - d), 0\}. \quad (71)$$

Estate-tax revenue is

$$\mathcal{R}_t^b = \tau_t^b \int E_t(s) dD_t(s), \quad (72)$$

where  $D_t$  is the decedent distribution. After-tax estates are assigned to heirs using a kernel  $\Gamma_t(s^h \mid s^d)$ :

$$B_{t+1}(s^h) = \int (1 - \tau_t^b) E_t(s^d) \Gamma_t(s^h \mid s^d) dD_t(s^d). \quad (73)$$

A family-linked kernel assigns estates to actual descendants; an anonymous kernel draws heirs from the aggregate estate distribution. Estate-tax revenue is rebated according to  $T_t^b(s)$  satisfying

$$\int T_t^b(s) d\mu_t(s) = \mathcal{R}_t^b \quad (74)$$

when the reform is revenue-neutral. The schedule  $T_t^b$  is the estate-revenue component of total transfers  $T_t(s)$  in the government budget. A repeal counterfactual must specify both  $\Gamma_t$  and  $T_t^b$ .

Old households may also have true warm-glow bequest utility

$$\mathcal{B}(E_t) = \theta_B \frac{[(1 - \tau_t^b)E_t + \underline{b}]^{1-\sigma_B}}{1 - \sigma_B}. \quad (75)$$

This is not an inefficiency. The planner counts (75). It is distinct from policy-created retention wedges.

## 11 Housing supply and units

For each size  $k \in \mathcal{K}$ ,  $H_{k,t}$  is measured in service units. A size- $k$  dwelling delivers  $h_k$  services. The stock law is

$$H_{k,t+1} = (1 - \delta_k)H_{k,t} + I_{k,t}, \quad (76)$$

where  $I_{k,t}$  is construction in service units. Developers have cost

$$C_{k,t}(I) = \frac{\chi_k}{1 + \eta_k} \left( \frac{I}{Z_{k,t}} \right)^{1+\eta_k} Z_{k,t}, \quad \eta_k > 0. \quad (77)$$

Competitive construction implies

$$P_{k,t} = C'_{k,t}(I_{k,t}) = \chi_k \left( \frac{I_{k,t}}{Z_{k,t}} \right)^{\eta_k}. \quad (78)$$

The rental/user cost satisfies

$$R_{k,t} = (r_t + \delta_k + \tau_t^p) P_{k,t} - \mathbb{E}_t(P_{k,t+1} - P_{k,t}). \quad (79)$$

Competitive landlords own rental units and rent them to households. Let  $H_{k,t}^O$  be owner-occupied services and  $H_{k,t}^R$  landlord-owned rental services. Landlords are risk-neutral price takers; zero profit gives (79), and rental supply satisfies  $H_{k,t}^R = H_{k,t} - H_{k,t}^O$ .

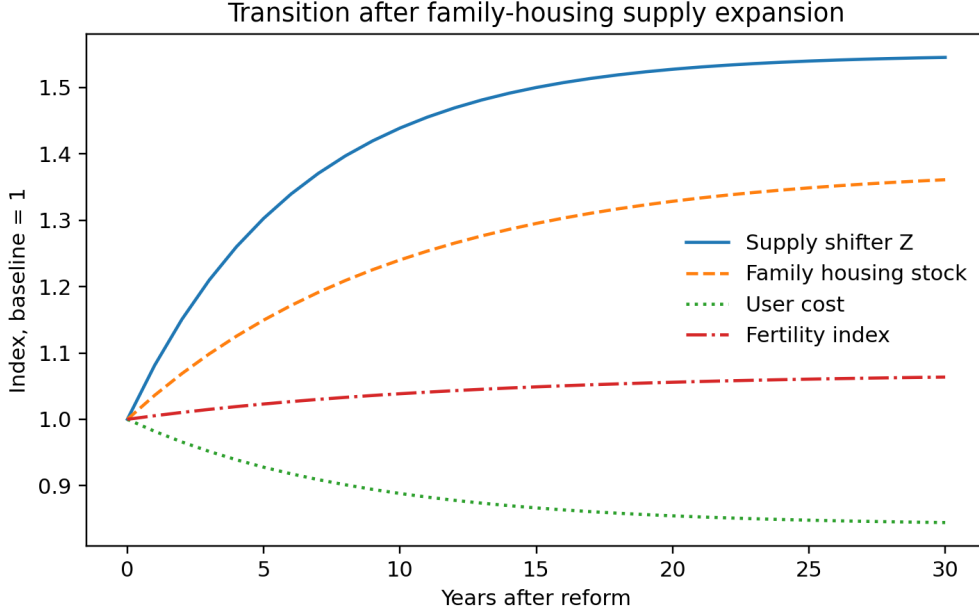


Figure 5: Schematic: supply expansion in the quantitative model. A rise in  $Z_{M,t}$  or  $Z_{L,t}$  increases construction of family-capable housing and reduces user costs along the transition.

## 12 Household dynamic program

Let  $V_t(s)$  be the household value. With  $n' = n + f_t$  and  $g'$  determined by (58),

$$V_t(s) = \max_{c, b', o', k', d', m^{d'}, l', f_t} \{u(c, k', n', g') + \beta \pi_a \mathbb{E}[V_{t+1}(s') \mid s, \text{choices}] + \beta(1 - \pi_a) \mathcal{B}(E_{t+1}(s'))\} \quad (80)$$

subject to (70), (65)–(66), income transitions, child aging, tenure feasibility, mortgage amortization, bequest transmission, and nonnegativity constraints. The next state is

$$s' = (a + 1, z', b', o', k', d', m^{d'}, l', n', g', B'). \quad (81)$$

## 13 Recursive competitive equilibrium

Let  $\mu_t$  be the distribution over states. A recursive competitive equilibrium is a sequence of prices  $\{P_{k,t}, R_{k,t}, R_t^m, r_t\}$ , taxes and transfers, construction  $\{I_{k,t}\}$ , policies, and distributions  $\{\mu_t\}$  such that:

1. Households solve (80).
2. The distribution evolves under policies, survival, income shocks, sequential births, child aging, deaths, bequests, and inheritance transitions.
3. Housing markets clear by size and tenure in service units. Owner-occupied demand and rental demand are

$$\begin{aligned} H_{k,t}^O &= \int \mathbf{1}\{o_t(s) = O, k_t(s) = k\} h_k d\mu_t(s), \\ H_{k,t}^R &= \int \mathbf{1}\{o_t(s) = R, k_t(s) = k\} h_k d\mu_t(s). \end{aligned} \tag{82}$$

The total stock clears:

$$\boxed{H_{k,t}^O + H_{k,t}^R = H_{k,t}, \quad k \in \mathcal{K}.} \tag{83}$$

Landlords supply  $H_{k,t}^R$  and satisfy the zero-profit condition (79).

4. Stocks evolve by (76) and construction satisfies (78).
5. The government and policy-wedge accounts clear:

$$\tau_t^p \sum_{k \in \mathcal{K}} P_{k,t} H_{k,t} + \mathcal{R}_t^b + \mathcal{R}_t^{other} = \int T_t(s) d\mu_t(s) + G_t + \mathcal{C}_t^L, \tag{84}$$

where  $\mathcal{C}_t^L$  is the fiscal, lender, or tax-expenditure cost of any explicit policy-created retention wedge. If lock-in is purely a private mortgage-contract transfer, set  $\mathcal{C}_t^L = 0$  and book the offset on the lender side.

6. Mortgage and bond markets clear, or  $r_t$  and  $R_t^m$  are taken from an external credit market with an exogenous spread.

## 14 Planner and welfare

The planner uses the same technology. Goods feasibility includes construction:

$$\int c_t(s) d\mu_t(s) + \sum_{k \in \mathcal{K}} C_{k,t}(I_{k,t}) + \text{child costs} + \text{maintenance} = Y_t. \tag{85}$$

Let  $Q_{k,t}$  be the shadow value of one service unit of size  $k$ . The planner supply condition is

$$Q_{k,t} = C'_{k,t}(I_{k,t}) + \beta_t(1 - \delta_k)\mathbb{E}_t Q_{k,t+1}. \tag{86}$$

The planner counts true warm-glow bequest utility. It does not count policy-created retention wedges as social benefits unless they reflect true preferences or real resource costs.

## 15 Calibration and required quantitative output

A complete quantitative section needs a parameter vector, model fit, and counterfactual results. The following tables are scaffolds and should not be read as calibrated estimates.

Table 3: Parameter blocks to calibrate

| Block       | Parameters                                                  | Main targets                                                                                 |
|-------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Preferences | $\beta, \sigma_u, \alpha_h, \varphi_n, \kappa, \chi_0, \xi$ | wealth profile, tenure by age, housing by children, completed fertility, age-specific births |
| Income      | $e_a, \Pi_z, W_t$                                           | age-income profile, dispersion, persistence                                                  |
| Credit      | $\phi, \psi, R_t^m - r_t$                                   | origination LTVs, payment-to-income ratios, mortgage spreads                                 |
| Lock-in     | $\lambda_s, \mathcal{L}^m$                                  | mobility responses to rate gaps, refinancing, downsizing hazards                             |
| Supply      | $\chi_k, \eta_k, Z_{k,t}, \delta_k$                         | construction by size, price-rent ratios, supply elasticities                                 |
| Bequests    | $\tau^b, \Gamma_t, T_t^b, \theta_B$                         | estate sizes, inheritances, rebate incidence                                                 |

Table 4: Model-fit table template

| Moment                                     | Data | Model | Role                                |
|--------------------------------------------|------|-------|-------------------------------------|
| Homeownership by age and income            | TBA  | TBA   | Tenure and credit margins           |
| House size by age and child status         | TBA  | TBA   | Mismatch and family-housing demand  |
| Birth hazards by housing transition        | TBA  | TBA   | Fertility-housing elasticity        |
| Old downsizing hazard by mortgage-rate gap | TBA  | TBA   | Lock-in                             |
| Inheritance receipt by heir age and income | TBA  | TBA   | Bequest-tax incidence               |
| Construction elasticity by size            | TBA  | TBA   | Supply attenuation of price effects |

Table 5: Counterfactual results table template

| Policy                                 | $\Delta$ fertility | $\Delta$ young family housing | $\Delta$ old retention | Welfare CEV |
|----------------------------------------|--------------------|-------------------------------|------------------------|-------------|
| Property-tax reform with rebates       | TBA                | TBA                           | TBA                    | TBA         |
| Mortgage portability / lock-in relief  | TBA                | TBA                           | TBA                    | TBA         |
| Family-housing supply expansion        | TBA                | TBA                           | TBA                    | TBA         |
| Bequest-tax repeal, revenue-neutral    | TBA                | TBA                           | TBA                    | TBA         |
| Targeted inheritance housing exemption | TBA                | TBA                           | TBA                    | TBA         |
| Combined reform                        | TBA                | TBA                           | TBA                    | TBA         |

## 16 Counterfactual protocol

Policies should be evaluated as transitions. For each policy, report: direct household effects holding prices fixed; price effects with markets cleared; supply effects with construction responding; distribution effects from fertility, aging, deaths, and bequests; and the full transition. This is a simulation decomposition protocol, not an exact derivative identity.

The main counterfactuals are property-tax reform with rebates, mortgage portability or lock-in relief, family-housing supply expansion, bequest-tax repeal under explicit transfer rules, targeted inheritance exemptions for first-time family-housing purchases, and a combined reform.

## 17 Limitations and next steps

The model deliberately drops space. This buys clarity and computational tractability but removes local labor markets, amenities, school quality, and spatial substitution. Housing supply is national and size-specific, not local. Fertility is continuous in the compact model and sequential discrete in the quantitative model. The mortgage lock-in term is written as a present-value object, but implementation must track amortization, refinancing, and remaining maturity. Bequests are represented with a parsimonious kernel; real estate taxation in data includes estate taxes, capital-gains taxation, basis step-up, inter vivos transfers, and institutional details.

The next step is not to add space back. The next step is to discipline the non-spatial margins: young liquidity constraints, payment-to-income constraints, old lock-in and downsizing elasticities, construction elasticity of family-capable units, inheritance incidence, and the fertility response to housing access.

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